Mathematics A

Unit 1 • Chapter OL-T50 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 19 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T50 • Expertise Q16+ • 4 items • 19 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

3D Pythagoras & Trigonometry

Unit 1 • Expertise • 19 marks • 4 item(s)

January 2022 | 4MA1-1H Q21 | 5 marks

Use midpoint or symmetry structure in a 3D model

June 2021 | 4MA1-1H Q22 | 4 marks

Use a non-rectangular base before a line-plane angle

May-Nov 2020 | 2H Q18 | 5 marks

Find angle between a line and a plane

January 2023 | 2H Q23 | 5 marks

Use 3D trigonometry to obtain a terminal volume

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q21 **5 marks**

21 The diagram shows the prism *ABCDEFGHJK* with horizontal base *AEFG*

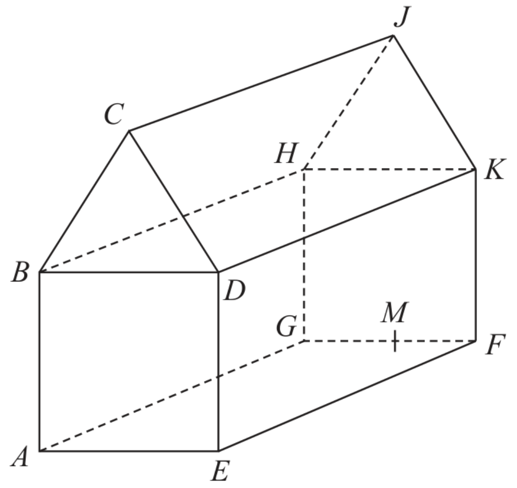


Diagram **NOT** accurately drawn

ABCDE is a cross section of the prism where

ABDE is a square

BCD is an equilateral triangle

$EF = 2 \times AE$

M is the midpoint of *GF* so that *JM* is vertical.

Angle $MAJ = y°$

Given that $\tan y° = T$

find the value of *T*, giving your answer in the form $\dfrac{\sqrt{p}+\sqrt{q}}{17}$ where *p* and *q* are integers.

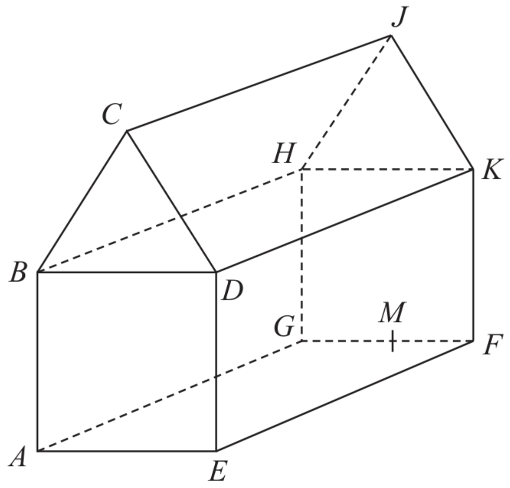
$T =$ ....

Continue your method **5 marks**

WORKING SPACE

4MA1-1H Q21 **5 marks**

21 The diagram shows the prism *ABCDEFGHJK* with horizontal base *AEFG*



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ABCDE is a cross section of the prism where
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BCD is an equilateral triangle

$EF = 2 \times AE$
M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y°$

Given that $\tan y° = T$

find the value of T, giving your answer in the form $\dfrac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T =$ ....

Chapter: 3D Pythagoras & Trigonometry **5 marks**

Q21 – Chapter: 3D Pythagoras & Trigonometry

Family: Resolve multi-stage solid geometry • Variant: Use midpoint or symmetry structure in a 3D model

Method
$AM=\sqrt{(x^2+(4x)^2)}=x\sqrt{17}$.
Why: Use Pythagoras in the horizontal base triangle.

Method
The equilateral-triangle height is $\sqrt{((2x)^2-x^2)}=x\sqrt{3}$.
Why: Use the altitude of the equilateral cross-section.

Method
$\tan y=(x\sqrt{3}+2x)/(x\sqrt{17})=(\sqrt{3}+2)/\sqrt{17}$.
Why: Use opposite over adjacent for angle MAJ.

Method
Rationalising gives $(\sqrt{3}+2)\sqrt{17}/17=(\sqrt{51}+2\sqrt{17})/17$.
Why: Multiply numerator and denominator by $\sqrt{17}$.

Method
Since $2\sqrt{17}=\sqrt{68}$, $T=(\sqrt{51}+\sqrt{68})/17$.
Why: Write the answer in the requested surd form.

Final answer
T: $(\sqrt{51}+\sqrt{68})/17$

22 The diagram shows a triangular prism *ABCDEF* with a horizontal base *ABEF*.

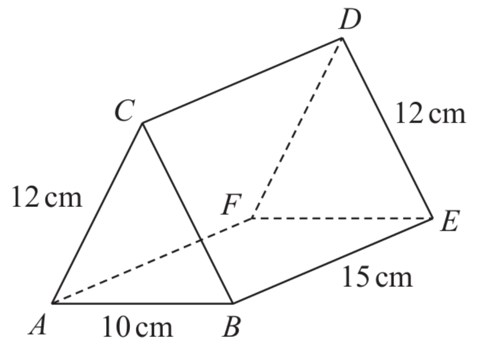

Diagram **NOT** accurately drawn

$AC = BC = FD = ED = 12\text{ cm}$ $\qquad AB = 10\text{ cm}$ $\qquad BE = 15\text{ cm}$

Calculate the size of the angle between *AD* and the base *ABEF*.
Give your answer correct to 3 significant figures.

....°

Worked solution

June 2021 | Question 22 | 4 marks

Family: Resolve multi-stage solid geometry • Variant: Use a non-rectangular base before a line-plane angle

Method / working

1. Use the prism diagram to form the right triangles with lengths 15.8, 19.2 and 10.9 cm.
2. Use tan(angle)=10.9/15.8 (equivalently a sine/cosine relation).
3. Evaluate the inverse trigonometric function.
4. Round to 1 decimal place: 34.6°.

Final answer: 34.6°

18 The diagram shows cuboid *ABCDEFGH*.

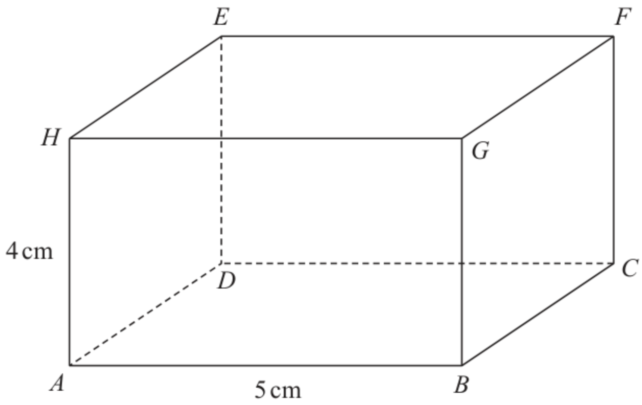

Diagram **NOT** accurately drawn

$AB = 5\,\text{cm}$
$AH = 4\,\text{cm}$
The size of the angle between *CH* and the plane *ABCD* is 35°

Calculate the volume of the cuboid.
Give your answer correct to 3 significant figures.

.... cm^3

(Total for Question 18 is 5 marks)

Answer reference | May-Nov 2020 | Question 18

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

18	eg $\frac{4}{AC} = \tan 35$ oe **or** $\frac{AC}{4} = \tan 55$ oe **or** $\frac{AC}{\sin 55} = \frac{4}{\sin 35}$ oe **or** $CH = \frac{4}{\sin 35}$ oe (= 6.97...) **and** $\frac{AC}{"6.97"} = \cos 35$ oe **or** $CH = \frac{4}{\sin 35}$ oe (=6.97...) **and** $AC^2 = 6.97^2 - 4^2$ oe			M1	A correct trig statement involving AC **or** trig and then Pythagoras involving AC
	$(AC =) \frac{4}{\tan 35}$ oe eg $(AC =)$ 4tan55 (= 5.71...) **or** $(AC =) \frac{4 \sin 55}{\sin 35}$ **or** "6.97" × cos35 oe **or** $(AC =)$ $\sqrt{"6.97"^2 - 4^2}$			M1	complete method to find AC
	$(BC =)$ $\sqrt{"5.71"^2 - 5^2}$ (= 2.76...)			M1	complete method to find BC
	4 × 5 × "2.76..."			M1	method to find volume
		55.3	5	A1	accept 55.1 – 55.5
					Total 5 marks

Worked solution

May-Nov 2020 | Question 18 | 5 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Official final mark-scheme pages are embedded immediately after this question.

23 The diagram shows a solid prism *ABCDEFGHIJ*

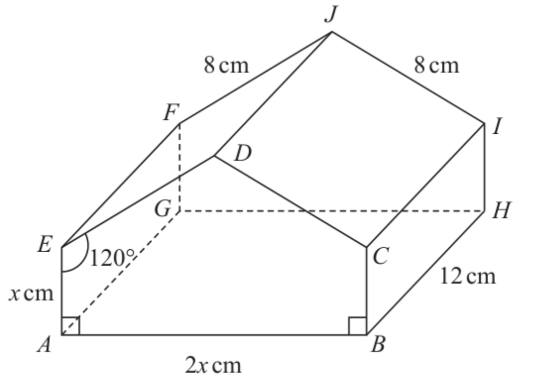

Diagram **NOT** accurately drawn

The prism is such that each cross section is a pentagon where

$AE = BC = x$ cm $\qquad AB = 2x$ cm $\qquad ED = CD = 8$ cm

angle EAB = angle $CBA = 90°$ $\qquad$ angle AED = angle $BCD = 120°$

Given that $AG = BH = EF = DJ = CI = 12$ cm

calculate the angle that AJ makes with the base $ABHG$ of the prism.
Give your answer correct to 3 significant figures.

January 2023 | 2H | Question 23 | 5 marks

Chapter OL-T50: 3D Pythagoras & Trigonometry

°

....

(Total for Question 23 is 5 marks)

Answer reference | January 2023 | Question 23

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

Q	Working	Answer	Mark	Notes
23	$[DN =]8\sin 30$ **or** $8\cos 60$ $(= 4)$ oe [where N is the midpoint of EC] **or** $[x =]8\cos 30$ **or** $8\sin 60 (= 4\sqrt{3} = 6.928...)$ **or** $2x = \sqrt{8^2 + 8^2 - 2\times 8\times 8\times \cos 120}(= \sqrt{192} = 8\sqrt{3} = 13.85....)$		5	M1
	$[DN =]8\sin 30$ **or** $8\cos 60$ $(= 4)$ oe eg $\sqrt{8^2 - (4\sqrt{3})^2}(= 4)$ **And 1 of** $[x =]8\cos 30$ **or** $8\sin 60 (= 4\sqrt{3} = 6.928...)$ oe or $\sqrt{8^2 - "4"^2}(= 4\sqrt{3} = 6.928...)$ **or** $2x = \sqrt{8^2 + 8^2 - 2\times 8\times 8\times \cos 120}(= 8\sqrt{3} = 13.85....)$			M1 [$(JM =)$ $4 + 4\sqrt{3}$ implies M2]
	$[AM =]\sqrt{12^2 + ("4\sqrt{3}")^2}$ $(= \sqrt{192} = 8\sqrt{3} = 13.856\ldots)$ oe [where M is the midpoint of GH]			M1 **Clear intention to be AM (not EC)**
	$\tan MAJ = \left(\frac{"4"+"4\sqrt{3}"}{"8\sqrt{3}"}\right)$ oe eg $\tan MAJ = \left(\frac{10.928...}{13.856...}\right)$ if student uses sin or cos, then $AJ = 17.647\ldots$ to award marks this must come from a correct method or be correct			M1
	Correct answer scores full marks (unless from obvious incorrect working)	38.3		A1 Accept 38.1 – 38.4 If no marks scored then award SCB2 for $\tan MAJ = \frac{4+x}{\sqrt{x^2+12^2}}$ or SCB1 for $AM = \sqrt{x^2+12^2}$
				Total 5 marks

Worked solution

January 2023 | Question 23 | 5 marks

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

Official final mark-scheme pages are embedded immediately after this question.